

Pencil Code User Meeting 2026

PCUM-26 · 21–25 September 2026 · University of Alicante

All times local (CEST, UTC+2)

	Monday 21 <i>getting started & GPUs</i> <i>collecting coding topics</i>		Tuesday 22 <i>community & development</i>	Wednesday 23 <i>chiral MHD & dynamos</i>	Thursday 24 <i>numerical dissipation & PINN</i>	Friday 25 <i>final talks & wrap-up</i>
09:30 – 10:15	Introductions 2 min per person		Steering Committee Report	Deepen Garg	Simon Candelaresi	Alberto Roper Pol
10:15 – 11:00	Axel Brandenburg		Philippe Bourdin	Simon Candelaresi	Jorge Urban	Ramkishor Sharma <i>online</i>
11:00 – 11:30	Coffee		Coffee + group photo	Coffee	Coffee	Coffee
11:30 – 13:00	Tutorial: beginners Touko Puro	Tutorial: admin Philippe Bourdin	<i>Coding</i> Latest developments & GPU Axel & Touko	<i>Coding</i> Persistent variables Storing large arrays	<i>Coding</i> Finding pencils Python & IDL routines	Brainstorming GR in the Pencil Code Matthias Rheinhardt
13:00 – 15:00	LUNCH		LUNCH	LUNCH	LUNCH	FAREWELL LUNCH
15:00 – 15:45	Tutorial: GPU Touko Puro		EXCURSION	Clara Dehman	Vinay Kumar <i>online</i>	Concluding remarks + next year's agenda
15:45 – 16:30	<i>Coding</i> GPU			<i>Coding</i> Symplectic solver	<i>Coding</i> Numerical dissipation	FREE TIME
16:30 – 17:00	Coffee			Coffee	Coffee	
17:00 – 18:00	Neco Kriel			<i>Coding</i> Subcycling Magic and missing ipencils	<i>Coding</i> Spectral output	
<i>evening</i>	Welcome drinks informal				Conference dinner	

Talks

Axel Brandenburg – Pencil Code since 2025; admin tutorial (Mon)

Philippe Bourdin – Recent topics in Pencil Code development

Alberto Roper Pol – *Relativistic MHD in an expanding Universe with the Pencil Code*

Touko Puro – tutorials: beginners (Mon am), GPU (Mon pm)

Deepen Garg – Chiral MHD in Pencil Code

Clara Dehman – Neutron star physics

Ramkishor Sharma (online) – Magnetogenesis from a pseudoscalar–gauge field interaction

Jorge Urban – Hydrodynamics with physics-informed neural networks

Neco Kriel – the Pencil and Quokka codes for small-scale dynamos

Simon Candelaresi – Numerical dissipation in CFD codes (Wed). Multiphysics coupling (Thu).

Vinay Kumar (online) – Does the plasmoid-mediated reconnection rate depend on the magnetic Prandtl number?

Coding topics

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Collected on Monday and worked on in the collaborative coding sessions through the week.

Several topics were taken up by parallel groups within the same session, so not all of them appear separately in the programme.

- **Python and IDL routines**
- **Spectral output**
- **Persistent variables**
- **GPU**
- **Symplectic solver**
- **Subcycling**
- **Finding pencils**
- **Numerical dissipation**
- **Magic and missing ipencils**
- **Storing large arrays**
- **GR in the Pencil Code**